



**Introduction to
Internet of Things
Assignment-Week 7**

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15

Total marks: 15 X 1= 15

QUESTION 1:

In the remote data logging system, which socket type is used to send sensor data from the client to the server?

- a. SOCK_STREAM
- b. SOCK_DGRAM
- c. SOCK_RAW
- d. SOCK_SEQPACKET

Correct Answer: b. SOCK_DGRAM

Detailed Solution: The client creates a UDP socket using `socket.socket(socket.AF_INET, socket.SOCK_DGRAM)` to send the sensor readings to the server.

See Lecture 31 (Implementation of IoT with Raspberry Pi – Part 2) @ 07:10

QUESTION 2:

Which Adafruit function is used in the client code to read humidity and temperature values from the DHT sensor?

- a. `Adafruit_DHT.read_retry()`
- b. `Adafruit_DHT.get_data()`
- c. `Adafruit_DHT.sense()`
- d. `Adafruit_DHT.capture()`



Correct Answer: a. `Adafruit_DHT.read_retry()`

Detailed Solution: The `sensordata()` function calls `Adafruit_DHT.read_retry(sensor,17)`, which returns the humidity and temperature values from the DHT sensor.

See Lecture 31 (Implementation of IoT with Raspberry Pi – Part 2) @ 04:50

QUESTION 3:

Where does the server in the remote data logging example save the data received from the client as mentioned in the lecture?

- a. In a MySQL database
- b. In a text file named `Datalog.txt`
- c. On a cloud dashboard
- d. In an Excel spreadsheet

Correct Answer: b. In a text file named `Datalog.txt`

Detailed Solution: The server code opens '`Datalog.txt`' in append mode and writes the received message from the client into this text file.

See Lecture 31 (Implementation of IoT with Raspberry Pi – Part 2) @ 15:40

QUESTION 4:

Which Python string function is used to split the comma-separated sensor data into separate values on the server?

- a. `split()`
- b. `divide()`
- c. `partition()`



d. `separate()`

Correct Answer: a. `split()`

Detailed Solution: The `split()` function splits a string into multiple strings based on a specified separator/delimiter, e.g. `Data.split(",")` splits data wherever a comma is found.

See Lecture 32 (Implementation of IoT with Raspberry Pi – Part 3) @ 08:40

QUESTION 5:

State whether the following statement is True or False.

Statement: Matplotlib Python library is used to plot the temperature and humidity data received at the server in the lecture.

a. False

b. True

Correct Answer: b. True

Detailed Solution: MATPLOTLIB is a Python library used to create 2D plots; functions such as `plot()`, `xlabel()`, `ylabel()`, and `title()` are used to plot and label the sensor data.

See Lecture 32 (Implementation of IoT with Raspberry Pi – Part 3) @ 12:15

QUESTION 6:

Which matplotlib function turns on interactive mode so that the plot updates in real time as new data arrives?

a. `ion()`

b. `grid()`

c. `subplot()`

d. `figure()`



Correct Answer: a. `ion()`

Detailed Solution: `ion()` turns on the interactive mode in matplotlib. Other common plotting functions mentioned include `figure()`, `grid()`, `subplot()`, `close()`, and `scatter()`.

See Lecture 32 (Implementation of IoT with Raspberry Pi – Part 3) @ 23:25

QUESTION 7:

In a traditional (non-SDN) network, where does the routing protocol logic (e.g., OSPF) mainly execute?

- a. In a centralized controller
- b. At the individual switches
- c. On the end-user device
- d. In the cloud

Correct Answer: b. At the individual switches

Detailed Solution: In the current network, protocols like OSPF execute at the switches, which forward traffic in a distributed manner with no centralized control.

See Lecture 33 (Software-Defined Networking – Part I) @ 04:10

QUESTION 8:

In which year was the OpenFlow project — from which the idea of software-defined networking originated — presented at ACM SIGCOMM?

- a. 2006
- b. 2008
- c. 2009
- d. 2011



Correct Answer: b. 2008

Detailed Solution: The idea of software-defined networking originated from the OpenFlow project, presented at ACM SIGCOMM in 2008. Stanford published the OpenFlow V1.0.0 specification the following year.

See Lecture 33 (Software-Defined Networking – Part I) @ 10:50

QUESTION 9:

What message does a switch send to the SDN controller when it receives a packet for which no matching flow-rule exists?

- a. PACKET-IN message
- b. FLOW-REQUEST message
- c. RULE-MISS alert
- d. CONTROL-SYNC message

Correct Answer: a. PACKET-IN message

Detailed Solution: When no flow-rule is present for an incoming request, the switch sends a PACKET-IN message to the controller, which then decides a suitable flow-rule and inserts it at the switch.

See Lecture 33 (Software-Defined Networking – Part I) @ 22:33

QUESTION 10:

Which SDN API is used for communication between the control layer and the infrastructure layer?

- a. Northbound API
- b. Southbound API
- c. East-Westbound API
- d. North-Southbound API



Correct Answer: b. Southbound API

Detailed Solution: The Southbound API is used to communicate between the control layer and the infrastructure layer using the OpenFlow protocol. The Northbound API communicates between the control and application layers.

See Lecture 34 (Software-Defined Networking – Part II) @ 02:00

QUESTION 11:

According to the lecture, approximately how many requests per second can a typical SDN controller handle through a single thread?

- a. 50
- b. 100
- c. 200
- d. 500

Correct Answer: c. 200

Detailed Solution: The lecture states that a controller can typically handle around 200 requests in a second through a single thread.

See Lecture 34 (Software-Defined Networking – Part II) @ 04:10

QUESTION 12:

Which control mechanism is described where the entire network's control decisions are taken by a single controller?

- a. Distributed control
- b. Centralized control
- c. Hierarchical control



d. Federated control

Correct Answer: b. Centralized control

Detailed Solution: In centralized control, control decisions are taken in a centralized manner by a single controller, unlike distributed control where each subnetwork is controlled by a different controller.

See Lecture 34 (Software-Defined Networking – Part II) @ 08:15

QUESTION 13:

Which of the following is listed as a benefit of integrating SDN with IoT?

- a. Increased hardware cost
- b. Visibility of network resources
- c. Reduced sensor accuracy
- d. Elimination of all controllers

Correct Answer: b. Visibility of network resources

Detailed Solution: Benefits of integrating SDN in IoT include intelligent routing decisions, simplified information collection/analysis/decision-making, visibility of network resources, and intelligent traffic pattern analysis.

See Lecture 35 (Software-Defined IoT Networking – Part I) @ 02:50

QUESTION 14:

According to the lecture, what is a key resource-constraint challenge in wireless sensor networks that motivates software-defined WSN?

- a. Excess bandwidth
- b. Limited memory for control programs



- c. Too many controllers
- d. Unlimited computation power

Correct Answer: b. Limited memory for control programs

Detailed Solution: Wireless sensor network challenges include real-time programming of sensor nodes, vendor-specific architecture, resource-constrained (heavy computation cannot be performed) devices, and limited memory, meaning too many control programs cannot be inserted.

See Lecture 35 (Software-Defined IoT Networking – Part I) @ 08:10

QUESTION 15:

In the SDN-WISE protocol stack for sensor nodes, what does "INPP" stand for?

- a. Integrated Node Path Protocol
- b. Internal Node Power Processing
- c. In Network Packet Processing
- d. IoT Network Priority Protocol

Correct Answer: c. In Network Packet Processing

Detailed Solution: In the SDN-WISE protocol stack, the forwarding layer above IEEE 802.15.4 consists of flow-rules, and INPP stands for In Network Packet Processing.

See Lecture 35 (Software-Defined IoT Networking – Part I) @ 15:35

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